

After years of traditional computing...

I finally step into a new world.

A world where information can be more than just 0 or 1.

This is...
Quantum Computing!

**Different rules.
Bigger possibilities.**

Welcome, Student.
Before we meet qubits,
you must understand
the idea of a
computing model.

Sensei Q
Mentor

A computing model?
Is it just a type of
computer?

Not just a computer...

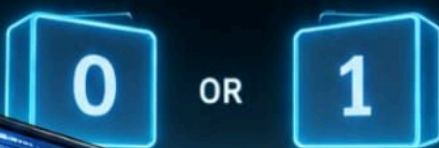
A computing model is a framework that defines:

1. How information is represented
2. How it is processed
3. How we obtain the result

Classical computing and quantum computing follow different models with different rules.

CLASSICAL COMPUTING MODEL

Uses bits to represent information.
Follows definite rules.



- ☒ Information: Bits (0 or 1)
- ☒ Processing: Logic Gates
- ☒ Output: Definite result
- ☒ Example: Laptops, Phones

QUANTUM COMPUTING MODEL

Uses qubits to represent information.
Follows quantum rules.



- ☒ Information: Qubits (superposition)
- ☒ Processing: Quantum Gates
- ☒ Output: Probabilistic result
- ☒ Example: Quantum Computers

So the key difference
is how information is
represented and
processed...

Exactly!
And that starts with
the qubit.

NEXT:
LET'S MEET THE QUBIT. >>

I know a classical bit can be 0 or 1...
But what exactly is a qubit?

A qubit is the basic unit of quantum information.

Unlike a classical bit, it can exist in a superposition of 0 and 1.

$|0\rangle$ $|1\rangle$

QUBIT

More than 0 or 1.

SAME
INFORMATION.
BIGGER
POSSIBILITIES.

QUANTUM
ACADEMY

A CLASSICAL BIT

A classical bit can be in only one state at a time: 0 or 1.

0

State: 0
(OFF)

1

State: 1
(ON)

Classical computers use many bits to process information following fixed rules.

A QUANTUM QUBIT

A qubit can be in a superposition of 0 and 1 at the same time.

$|0\rangle$ $|1\rangle$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Here, α and β are complex numbers called probability amplitudes.

$$|\alpha|^2 + |\beta|^2 = 1 \quad (\text{normalization})$$

So a qubit isn't just 0 or 1...
It's both, until we measure it?

QUESTION
EXPLORE
UNDERSTAND
GROW

Exactly!

A qubit exists in a superposition, and when we measure it, we get either 0 or 1 with certain probabilities.

Hi! I'm Qubi!
I can be in multiple states at once!

When you measure me, I collapse to either 0 or 1.
But until then, I'm a superposition!

QUBI

MEASUREMENT

$|0\rangle$ $|1\rangle$

Before measurement
(superposition)

Measure

0

with probability $|\alpha|^2$

1

with probability $|\beta|^2$

After measurement, the qubit collapses to a definite state (0 or 1).

This is amazing...
A single qubit already has more possibilities than a classical bit!

I can't wait to learn more!

Great!
In the next panels, we'll explore basis states, Bloch sphere and how to visualize qubits.

For now, remember—a qubit is the foundation of quantum computing.

NEXT:
BASIS STATES >>

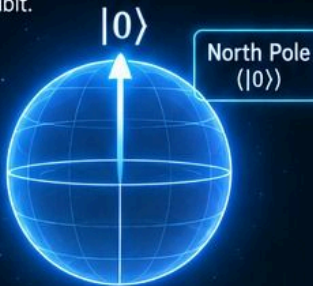
So a qubit can be in a superposition... But what are these $|0\rangle$ and $|1\rangle$ exactly?

Great question! $|0\rangle$ and $|1\rangle$ are called the **basis states** of a qubit. They are like the “north” and “south” poles of the qubit’s world.

BASIS STATES THE FOUNDATION OF A QUBIT

THE $|0\rangle$ STATE

$|0\rangle$ represents the ground state of a qubit.



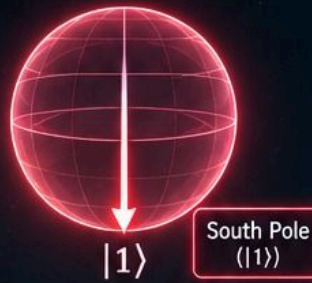
Mathematical representation:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

This is the state where the probability of measuring 0 is 100%.

THE $|1\rangle$ STATE

$|1\rangle$ represents the excited state of a qubit.



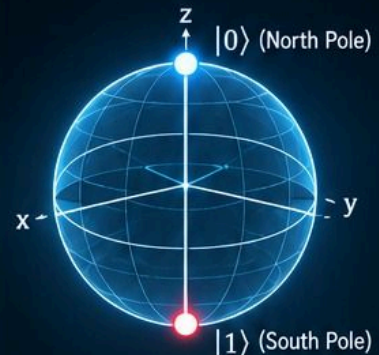
Mathematical representation:

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

This is the state where the probability of measuring 1 is 100%.

ON THE BLOCH SPHERE

The two basis states are at opposite poles on the Bloch sphere.



Any qubit state can be represented as a point on the surface of this sphere (but for now, remember the poles!).

So $|0\rangle$ and $|1\rangle$ are just like reference points... And any other qubit state is somewhere in between?

Exactly! They form a basis for the qubit’s state space.

QUICK CHECK

Which state is this?

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

A

$|0\rangle$

B

$|1\rangle$

Think before you answer!

Well done!

These basis states might look simple, but they are the building blocks of everything in quantum computing — gates, algorithms, and more.

“A deep understanding of simple things leads to extraordinary things.”

— Sensei Q

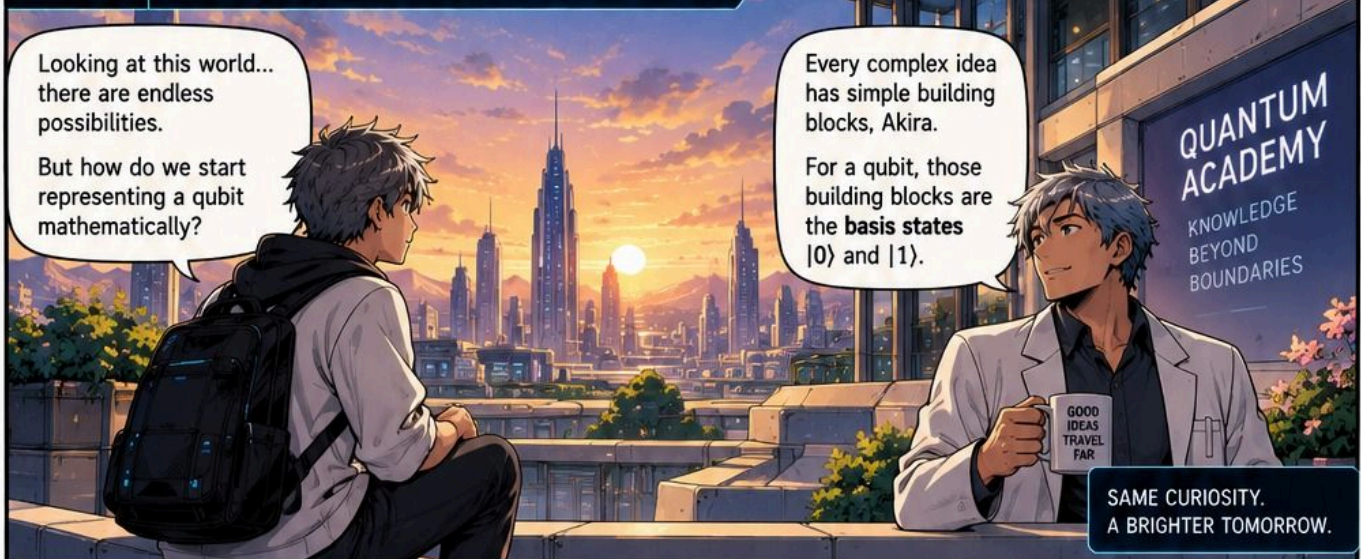
I started with a simple question... but now I see, these two states are the foundation of an entirely new world.

NEXT:
SUPERPOSITION



Same states.
More possibilities.

SMALL
CONCEPTS
BIG
REVOLUTIONS

THE $|0\rangle$ STATE — GROUND STATE

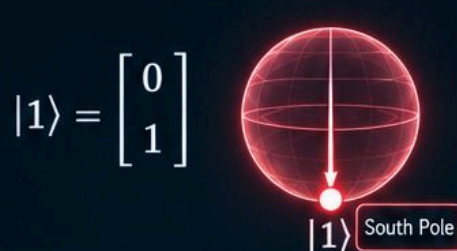
$|0\rangle$ is called the ground state. It represents the basic state of a qubit with 100% probability of measuring 0.



It's like the starting point of the qubit's journey.

THE $|1\rangle$ STATE — EXCITED STATE

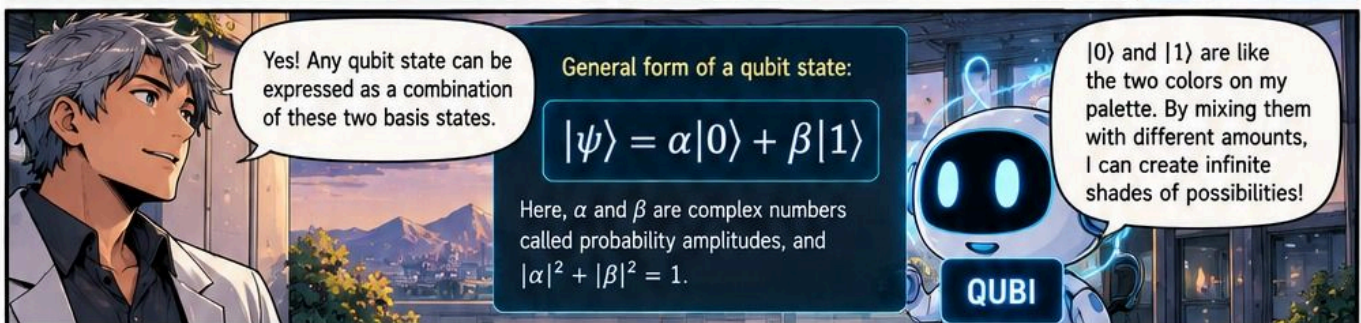
$|1\rangle$ is called the excited state. It represents the state with 100% probability of measuring 1.



It's another fundamental state, opposite to $|0\rangle$.

So $|0\rangle$ and $|1\rangle$ are just two reference points on the Bloch sphere? That's actually beautiful!

Simple ideas can lead to extraordinary worlds.



Wow...
The universe looks
so different from here.

It makes me feel like
there are infinite
possibilities out there.

JUST LIKE THE UNIVERSE,
A QUBIT CAN EXIST IN
MULTIPLE POSSIBILITIES
AT THE SAME TIME.

Exactly, Akira.
A qubit isn't limited to
0 or 1. It can be in a
superposition of both
— just like a star that
can take many paths
in space.

WHAT IS SUPERPOSITION?

A qubit can be in a combination
of $|0\rangle$ and $|1\rangle$ at the same time.
This is called superposition.



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Here, α and β are complex numbers
(probability amplitudes) and
 $|\alpha|^2 + |\beta|^2 = 1$.

A SIMPLE EXAMPLE

If $\alpha = 1/\sqrt{2}$ and $\beta = 1/\sqrt{2}$,
then the qubit is in an equal
superposition.

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

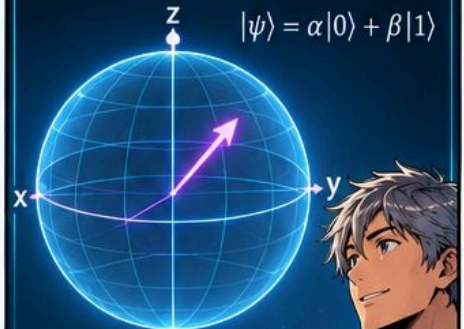


$P(0) = 50\%$
 $P(1) = 50\%$

So before we
measure, it's not just
0 or 1... it's both,
with certain
probabilities!

VISUALIZING SUPERPOSITION

A qubit in superposition can be represented
as a point on the Bloch Sphere.



Every point on the
surface represents
a different superposition
state. The poles are $|0\rangle$
and $|1\rangle$, and everything
in between is a valid
qubit state.

REAL-WORLD ANALOGY

Think of a spinning coin.
While it's spinning in the air,
it's not yet heads or tails —
it's a combination of both.

Only when it lands, we see
a definite result.

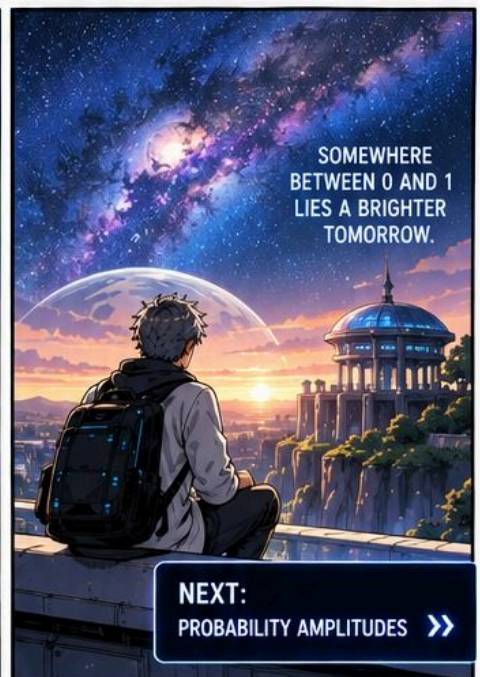


So superposition
is like a spinning coin...
but at the quantum level!

KEY TAKEAWAYS

- ✓ A qubit can be in a superposition
of $|0\rangle$ and $|1\rangle$.
- ✓ It is represented as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.
- ✓ α and β are complex probability
amplitudes.
- ✓ $|\alpha|^2 + |\beta|^2 = 1$.
- ✓ Measurement collapses the qubit
to 0 or 1.
- ✓ Superposition gives quantum
computers their massive power!

*"More than one state.
A world of possibilities."*



SOMEWHERE
BETWEEN 0 AND 1
LIES A BRIGHTER
TOMORROW.

NEXT:
PROBABILITY AMPLITUDES >>



PROBABILITY AMPLITUDES

A qubit state is written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

α and β are complex numbers called probability amplitudes.

The probabilities are given by the squared magnitudes:

$$P(0) = |\alpha|^2 \quad P(1) = |\beta|^2$$

$$|\alpha|^2 + |\beta|^2 = 1 \text{ (normalization)}$$

EXAMPLE 1 — EQUAL SUPERPOSITION

If $\alpha = 1/\sqrt{2}$ and $\beta = 1/\sqrt{2}$, then the qubit is equally likely to be 0 or 1.

State: $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$



$P(0) = 50\%$
 $P(1) = 50\%$



Wow!
So the qubit has an equal chance of giving 0 or 1 when measured!

EXAMPLE 2 — MORE 0 THAN 1

If $\alpha = \sqrt{0.8}$ and $\beta = \sqrt{0.2}$, then the qubit is more likely to be 0.

$$|\psi\rangle = \sqrt{0.8}|0\rangle + \sqrt{0.2}|1\rangle$$



$P(0) = 80\%$
 $P(1) = 20\%$

A larger amplitude for $|0\rangle$ means a higher probability of measuring 0.

EXAMPLE 3 — MORE 1 THAN 0

If $\alpha = \sqrt{0.3}$ and $\beta = \sqrt{0.7}$, then the qubit is more likely to be 1.

$$|\psi\rangle = \sqrt{0.3}|0\rangle + \sqrt{0.7}|1\rangle$$



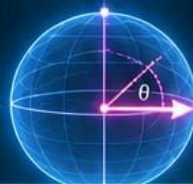
$P(0) = 30\%$
 $P(1) = 70\%$

A larger amplitude for $|1\rangle$ means a higher probability of measuring 1.

PHASE ALSO MATTERS

Amplitudes can be complex numbers. The phase (angle) affects how qubits interfere in algorithms.

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + e^{i\theta} \frac{1}{\sqrt{2}}|1\rangle$$



Different phases represent different points on the Bloch sphere, even if the probabilities are the same.

So α and β are not just normal numbers, they can have a direction (phase) too?

Exactly!
That's what makes quantum states rich and powerful. The phase plays a crucial role in interference, which we'll see in later chapters.



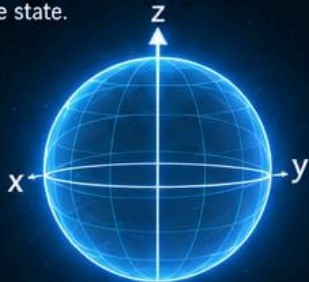
Woah...
What is this glowing
sphere, Sensei?

Is this a map of
all possible qubit
states?

Exactly, Akira!
This is the Bloch Sphere.
It's a powerful way to
visualize the state of a
single qubit. Every pure
qubit state can be
represented as a point
on the surface of this
sphere.

WHAT IS A BLOCH SPHERE?

A Bloch sphere is a geometric
representation of a single qubit's
pure state.



Any qubit state $|\psi\rangle$ can be written as:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

where:

θ = polar angle (0 to π)

ϕ = azimuthal angle (0 to 2π)

KEY POINTS ON THE SPHERE

$|0\rangle$ — North Pole

$$\theta = 0$$

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



North Pole
(0, 0, 1)

$|1\rangle$ — South Pole

$$\theta = \pi$$

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



South Pole
(0, 0, -1)

Superposition States

All other points on the surface
represent superposition states
of $|0\rangle$ and $|1\rangle$.

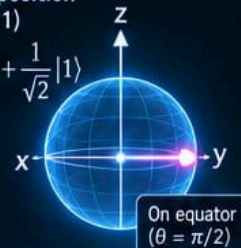


Superposition
(state between
 $|0\rangle$ and $|1\rangle$)

VISUALIZING DIFFERENT STATES

Equal Superposition
(50% 0, 50% 1)

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$



On equator
($\theta = \pi/2$)

Arbitrary Superposition

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$



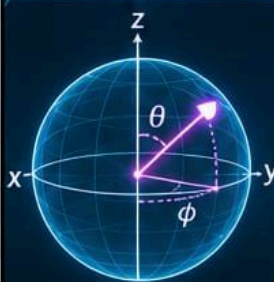
Point (θ, ϕ)
represents a
unique qubit state

Different angles
create different
superposition states.

So by just changing
the angles θ and ϕ ,
I can represent any
qubit state on this
sphere?

That's so intuitive!
It's like a coordinate
system for quantum
states!

INTERACTIVE VIEW (CONCEPT)



θ (polar angle)

$\pi/3$

ϕ (azimuthal angle)

$\pi/4$

Current State:

$$|\psi\rangle = \cos\left(\frac{\pi}{6}\right)|0\rangle + e^{i\frac{\pi}{4}}\sin\left(\frac{\pi}{6}\right)|1\rangle$$

Move the sliders to see how the point changes on the Bloch sphere.

Exactly, Akira.
The Bloch sphere makes
it easy to visualize qubit
states, understand
rotations (which we'll
explore in the next
chapter with gates),
and see how different
states relate to each
other.

Now I can actually see
the whole world of qubit states...
Not just 0 or 1, but an infinite
number of possibilities.

That's the power of the
Bloch sphere, Akira.
A single qubit contains
a universe of possibilities.

NEXT:
MEASURING A QUBIT >>

So we can represent qubit states... But how do we actually change a qubit from $|0\rangle$ to $|1\rangle$ or create superposition?

We use **quantum gates**. Gates are like operations that change the state of a qubit, just like classical gates (AND, OR, NOT) — but for quantum states.

QUANTUM GATES

H
X
Z
S
CNOT

QUANTUM GATES ARE THE TOOLS TO MANIPULATE QUBIT STATES.

QUANTUM RESEARCH LAB

EXPLORE
BUILD
EXPERIMENT

HADAMARD GATE (H)

Creates superposition from $|0\rangle$ or $|1\rangle$.



$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$



$|0\rangle \rightarrow$ superposition

PAULI-X GATE (X)

Flips the qubit state (like a classical NOT).



$$X|0\rangle = |1\rangle$$

$$X|1\rangle = |0\rangle$$



Rotates the state by 180° around the X-axis.

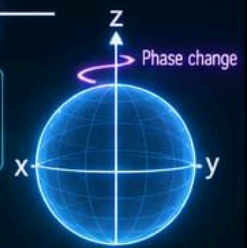
PAULI-Z GATE (Z)

Adds a phase of -1 to $|1\rangle$.



$$Z|0\rangle = |0\rangle$$

$$Z|1\rangle = -|1\rangle$$



Rotates the state by 180° around the Z-axis.

PHASE GATE (S)

Adds a phase of i to $|1\rangle$.



$$S|0\rangle = |0\rangle$$

$$S|1\rangle = i|1\rangle$$



Rotates the state by 90° around the Z-axis.

CNOT GATE (CONTROLLED-NOT)

Works on two qubits. Flips the target qubit if the control qubit is $|1\rangle$.



Control	Target	Output
0	0	00
0	1	01
1	0	11
1	1	10

Essential for creating entanglement and multi-qubit operations.

QUANTUM CIRCUIT EXAMPLE

Create a superposition, then measure.



Result after measurement:

$$P(|0\rangle) = 50\%$$

$$P(|1\rangle) = 50\%$$

So with just a few gates, we can create completely new quantum states!

Amazing... With quantum gates, we can create, rotate, combine and control qubit states.

These simple gates are the building blocks for complex algorithms like quantum teleportation, Shor's algorithm and more.

Exactly, Akira. Quantum gates give us the power to manipulate qubits, and when we combine them, we can solve problems that classical computers can't.

NEXT:
QUANTUM ENTANGLEMENT >>

We've learned that a qubit can be in a superposition of $|0\rangle$ and $|1\rangle$.

But what happens when we measure it, Akira?
Do we get both 0 and 1?

Good question!
When we measure a qubit, we get only one definite result — either 0 or 1.

The superposition collapses into a single state based on its probabilities.

MEASURE
OBSERVE
COLLAPSE
LEARN
BUILD

QUANTUM
COMPUTING
REAL WORLD
APPLICATIONS

1. BEFORE MEASUREMENT

The qubit is in a superposition state.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

For example, if $\alpha = 1/\sqrt{2}$ and $\beta = 1/\sqrt{2}$, the qubit has 50% chance of being 0 and 50% chance of being 1.



$P(0) = 50\%$
 $P(1) = 50\%$

2. DURING MEASUREMENT

Measurement collapses the superposition into one of the basis states.



Result:
0 or 1
(never both)

The probabilities decide which result we get.

3. AFTER MEASUREMENT

The qubit is now in a definite state.

If the result is 0:



$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$P(0) = 1$

If the result is 1:



$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$P(1) = 1$

4. MEASUREMENT RESULTS (MULTIPLE SHOTS)

If we prepare many identical qubits and measure them, we will see results according to the probabilities.

Example: $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$

Measure 1000 times
(approximately)



So if I measure many times, I don't always get the same result, but the distribution matches the probabilities!

5. REAL-WORLD ANALOGY

Think of a spinning coin.



While it's spinning, it's not clearly heads or tails — it's like a combination of both.

When it lands (we measure), we see only one result: heads or tails.

A qubit works in a similar way, but at the quantum level!

Now I understand!
Measurement gives a single result, but the real power comes from preparing, manipulating, and measuring many qubits together.

Exactly, Akira.
With measurement, we can extract useful information from quantum systems — which enables powerful algorithms and real-world applications.

NEXT:
WHY QUBITS MATTER >>

We can already do so much with classical computers.

So why do we need qubits? What makes them so special?

Great question! Qubits give us a completely new way to process information. They can be in multiple states at once, which opens the door to solving problems that classical computers struggle with.

QUANTUM
PEOPLE
IDEAS
REAL IMPACT

1. PARALLEL COMPUTATION

A classical bit can be either 0 or 1, so it processes one possibility at a time.

Classical (3 bits)

0 0 0



Checks 1 possibility at a time.

Quantum (3 qubits)



000 001 010 011
100 101 110 111

Checks all 8 possibilities at once (in superposition).

With n qubits, a quantum computer can represent 2^n states simultaneously!

2. FASTER SOLUTIONS

Quantum algorithms can solve certain problems much faster than classical algorithms.



Example:
Factoring large numbers
(Shor's Algorithm)

Could break encryption that keeps our data safe today.



Example:
Searching in large data
(Grover's Algorithm)

Finds the answer in $\sqrt{N}$ steps instead of N steps.

3. REAL-WORLD APPLICATIONS

Qubits enable solutions in many fields:



Drug Discovery

Simulate molecules and design new medicines.



Material Science

Create stronger, lighter, and more efficient materials.



Optimization

Solve complex logistics, scheduling and supply chain problems.



Artificial Intelligence

Speed up machine learning and data analysis.



Climate & Energy

Model climate systems and design better energy solutions.

4. BEYOND CLASSICAL LIMITS

Some problems are practically impossible for classical computers, but quantum computers can handle them efficiently.

Classical Computer
Tries many paths one by one.



Quantum Computer
Explores all paths at once.



5. A GLIMPSE OF THE FUTURE

Quantum computers are still in early stages, but they have the potential to revolutionize many industries.

Potential Impact:

- ☒ Safer communication
- ☒ Better healthcare
- ☒ Cleaner energy
- ☒ Smarter AI
- ☒ Solutions to global challenges

KEY TAKEAWAYS

- ☒ Qubits can be in multiple states at once (superposition).
- ☒ They enable parallel computation.
- ☒ Quantum algorithms solve certain problems much faster.
- ☒ Applications include healthcare, materials, optimization, AI, and climate.
- ☒ Qubits open up a new era of possibilities beyond classical limits.

"Qubits don't just compute faster — they help us solve bigger problems."

So qubits are not just a new type of bit...

They are the key to a future where we can solve problems that seem impossible today.

Exactly, Akira.

Qubits give us a new perspective on computation — and a brighter tomorrow for science, technology and society.

END OF CHAPTER 01

NEXT: CHAPTER 02 — QUANTUM CIRCUITS



So... I finally understand qubits now.
Their states, superposition,
probability amplitudes,
measurement, and even
the Bloch sphere...
It's amazing, Sensei!

Good, Akira.
You've learned how
a qubit behaves —
how to represent it,
how to measure it,
and why it's powerful.

But remember...
this is just the beginning.

A SMALL
QUBIT
TODAY,
A BRIGHTER
TOMORROW.

SUDDENLY...



WARNING
UNKNOWN QUANTUM STATE
DETECTED

What's happening?!
The academy systems
are showing an alert!

Unknown
quantum state?
But we just
measured it...

QUANTUM CORE
SIGNAL UNSTABLE
MEASUREMENT
RESULT INCONSISTENT
CHECK QUBIT ID: Q-001

Sensei...
We measured the qubit
and got 0 or 1.

How can its state
be changing again?

Look at the data,
Akira.

This behavior
doesn't match
our simulation.

THE IMPOSSIBLE QUBIT
QUBIT ID: Q-001

LIVE STATE MONITOR



$$\alpha = 0.71$$

$$\beta = 0.71$$

$$|\alpha|^2 = 0.50$$

$$|\beta|^2 = 0.50$$

But... this is impossible.
The results are not consistent.
It's not behaving like a normal qubit.

MEASUREMENT LOG

t = 00:00:01	0
t = 00:00:02	1
t = 00:00:03	0
t = 00:00:04	1
t = 00:00:05	?



ERROR

STATE DOES NOT
MATCH SIMULATION

No, Akira..
It means...
this isn't the qubit
we thought we were
studying.

They've learned
how to observe it.

Now let's see
if they can control it.

PROJECT: QUANTUM ZERO

SUBJECT Q-001
STATUS **AWAKENED**
PHASE 01 **COMPLETE**
NEXT **QUANTUM CIRCUITS**



CHAPTER 01 — END
CHAPTER 02: THE FIRST GATE

